

Q1. APRIORI ALGORITHMPseudo code:

START

Input transaction dataset

set minimum support

find all individual items

count support of each item

Remove items below minimum support

Keep the remaining items as frequent itemsets

WHILE frequent itemsets are available

Generate candidate itemsets

count support of each candidate

Remove candidates below minimum support

Keep the remaining candidates as frequent itemsets

END WHILE

Display all frequent itemsets

STOP.

Candidate Generation

From the dataset:

1. itemsets:

{ mobile }

{ earphones }

{ charger }

Generate:

2. itemsets:

{ mobile , earphones }

{ mobile , charger }

{ earphones , charger }

If the support is sufficient , generate:

3. itemsets:

{ mobile , earphones , charger }

②

FP - Growth Algorithm

START

Input borrowing transactions

Set minimum support

Count the frequency of each book

Remove books below minimum support

Sort remaining books by frequency

Create the FP-tree

Create the Header Table

For each book in the Header Table

Find its conditional pattern base

Create a conditional FP-tree

Generate frequent itemsets

END for

Display all frequent itemsets

STOP

FP - Tree construction

First Count the books.

python $\rightarrow 4$

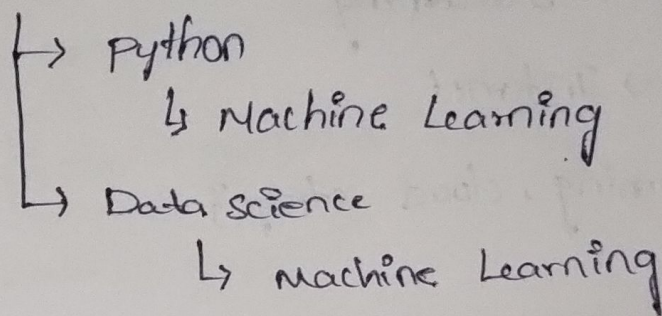
machine
Learning $\rightarrow 4$

Data science $\rightarrow 3$

Then arrange the books according to their frequency and
insert each transaction into the FP-Tree

For example:

Root



③

Association Rule Generation

Pseudo code

START

Input Frequent itemsets

Set minimum confidence

FOR each frequent itemset

Generate possible rules

For each possible Rules

Calculate confidence

IF confidence $\geq$ minimum confidence

Keep the rule

ELSE

Remove the rule

END IF

END FOR

END FOR

Display valid association rules

STOP

Rule Generation

From

{Internet, Streaming}

Possible rules

Internet $\rightarrow$ Streaming

Streaming $\rightarrow$ Internet

FROM: {Streaming, cloud storage}

Possible rules:

{ Streaming $\rightarrow$ cloud storage

cloud storage $\rightarrow$ streaming

Confidence:

Confidence tells us how often the second service is purchased when the first service is purchased.

Example:

Confidence (Internet $\rightarrow$ streaming)

$= \text{support (Internet, streaming)} / \text{support (Internet)}$

$= 3/4$

$\Rightarrow 75\%$

If the minimum confidence is 70%. Then this rule is accepted.

④

Constraint - Based Frequent Itemset mining

Dataset : student course Enrollment

Pseudocode:

START

Input student course transactions

set constraint = "python"

find all individual courses

Generate candidate itemsets

For each candidate itemsets

Generate candidate itemsets

For IF itemset contains "python"

calculate support

IF support $\geq$ minimum support

keep the itemset

ELSE

Remove the itemset

END IF

ELSE

Remove the itemset

END FOR

Generate largest itemsets containing "python"

Repeat until no more itemsets can be generated

Display frequent itemsets containing "python"

STOP

Candidate Generation

from the dataset:

{python}

{python, Databases}

{python, AI}

{ python, Database, AI }

only itemsets containing python are considered.

⑤ multilevel Association Rule mining

Dataset: Hospital Treatment Hierarchy

Pseudocode:

START

Input treatment transactions

Input treatment hierarchy

Set minimum support and confidence

Find frequent treatments at the top level

Generate frequent itemsets

For each hierarchy level

Generate frequent itemsets at that level

Calculate support

Remove itemsets below minimum support

Generate Association rules

Calculate confidence

Keep rules above minimum confidence

END For

STOP

Treatment Hierarchy

Healthcare

├── Cardiology
│ └── ECG
└── Orthopedics